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Research paper

Freezing and glass transitions upon cooling and warming and ice/freeze-concentration-solution morphology of emulsified aqueous citric acid

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ABSTRACT

Although freeze-induced phase separation and the ice/FCS (freeze-concentration solution) morphology of aqueous solutions play an important role in fields ranging from life sciences and biotechnology to geophysics and high-altitude ice clouds, their understanding is far from complete. Herein, using differential scanning calorimetry (DSC) and optical cryo-microscope (OC-M), we have studied the freezing and glass transition behavior and the ice/FCS morphology of emulsified 10–60 wt% CA (citric acid) solutions in the temperature region of ~308 and 153 K. We have obtained a lot of new result which are understandable and unclear. The most essential understandable results are as follows: (i) similar to bulk CA/H_2O, emulsified CA/H_2O also freezes upon cooling and warming and (ii) the ice/FCS morphology of frozen drops smaller than ~3–4 μm is less ramified than that of frozen bulk solutions. Unclear results, among others, are as follows: (i) in contrast to bulk solutions, which produce one freezing event, emulsified CA/H_2O produces two freezing events and (ii) in emulsions, drop concentration is not uniform. Our results demonstrate that DSC thermograms and OC-M images/movies are mutually supplementary and allow us to extract important information which cannot be gained when DSC and OC-M techniques are used alone.
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1. Introduction

In aqueous solution, citric acid (CA) forms strong hydrogen bonds between CA molecules themselves and with H_2O [1–3]. The hydrogen bonds between CA molecules are responsible for the formation of CA-nanoclusters [4–9] whose size and population increase with concentration. As temperature decreases, the CA-nanoclusters greatly contribute to increasing viscosity because their translational and rotational mobility is much smaller than that of unbound CA and H_2O molecules. CA is widely used in food industry [10] and as a buffer for the lyophilization (freeze-drying) of biopharmaceutical formulations [11,12], because of a broad buffering range, minimal pH changes with temperature, and its resistance to crystallization under typical freeze-drying conditions [13]. It is also used for the preparation of multicomponent excipients with increased T_g in order to reduce protein denaturation in lyophilized products [14,15]. Emulsified formulations containing CA are used in biotechnology, biopharmaceutics, tissue engineering, etc. For example, emulsions are used for the preparation of oil-based formulations as drug delivery systems [16] and porous scaffolds needed in biopharmaceutics and tissue engineering [17–19]. Three-dimensional (3D) porous scaffolds for tissue engineering are prepared from lyophilized double water/oil/water (W/O/W) emulsions [20]. Rapidly frozen and lyophilized W/O emulsions are also used as transdermal delivery carrier as a solid-in-oil nano-suspension [21].

Lyophilization consists of three main steps: the freezing step, which takes place upon cooling of formulations, and the primary and secondary drying steps which are performed upon subsequent warming [22–27]. During freezing, formulations/solutions can separate into pure ice and FCS which freezes/crystalizes or transforms to glass upon further cooling. This freeze-induced phase separation (FIPS) produces a specific for each formulation ice/FCS morphology which controls (i) resistance to the vapor flow of sublimated ice during the primary drying, (ii) formation of the porous medium of a resulting cake, and (iii) desorption of residual moisture from the porous cake during the secondary drying. Thus ice/FCS morphology formed during freezing controls the duration of lyophilization and quality attributes of final lyophilized products [22–31].

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Formulations are usually multicomponent. The knowledge of the freezing and glass transition behavior and the ice/FCS morphology of aqueous components is useful for the preparation of multicomponent formulations with more or less predicted low-temperature behavior and ice/FCS morphology. We recently reported about the low-temperature behavior and ice/FCS morphology of bulk CA/H_2O and sucrose/H_2O [32,33]. The main factors that influence FIPS and, consequently, ice/FCS morphology of bulk *binary* formulations are the concentration and molecular structure of solutes. Whether the ice/FCS morphology of frozen emulsified formulations differs from that of bulk formulations was not investigated before. What now is known is that the heterogeneous freezing temperature of *diluted* bulk solutions is much warmer than the homogeneous freezing temperature of emulsified micrometer-scaled drops. In contrast, bulk and emulsified *concentrated* solutions transform to glass at approximately similar T_g. Physically, the much colder freezing temperature of small drops may impact freezing rate and, consequently, ice/FCS morphology in comparison with that of bulk solutions. Another factor that may influence the ice/FCS morphology of emulsified drops is a surfactant/oil matrix which envelops drops in emulsions. To our best knowledge, there are few works which reported that a surfactant/oil matrix may influence the low-temperature behavior of aqueous drops [34]. Besides biotechnology, biopharmaceutics, etc., the knowledge of the low-temperature behavior and ice/FCS morphology of small CA/H_2O drops may be important for the atmosphere where aqueous drops containing CA have been observed [35].

In this paper, we present the results of the measurements of freezing and glass transition behavior and ice/FCS morphology of emulsified 10–60 wt% CA solutions. The measurements were performed using differential scanning calorimetry (DSC) and optical cryo-microscope (OC-M). The paper is structured as follows. In experimental Section 2, we describe DSC and OC-M measurements and in Section 3, we recall the main features of the ice/FCS morphology of bulk CA/H_2O. In Section 4, we rationalize the ice/FCS morphology of emulsified drops from the comparison of bulk and emulsion DSC thermograms and analyze OC-M images obtained *in situ* during the cooling/warming of emulsions. Conclusions are collected in Section 5.

2. Experimental

We prepared 10–60 wt% CA solutions by mixing >99% anhydrous citric acid (Merck) with the corresponding amount of ultrapure water. Solution-in-oil emulsions were produced according to a widely used emulsification technique described elsewhere [36–38]. As an oil-surfactant matrix we used a mixture of 80 wt% Halocarbon 0.8 oil (Halocarbon Products Corp.) and 20 wt% Lanolin (Sigma Aldrich) (thereafter the HL-matrix). Emulsions were prepared by magnetic stirring of the solution/HL-matrix mixtures of 1/10 by volume.

We investigated the freezing, melting and glass transition behavior of emulsified solutions with a Mettler Toledo DSC 822 calorimeter at the scanning cooling/warming rate of 3 K/min in the temperature range between ~308 and 153 K. The calibration of calorimeter and details about measurements are described elsewhere [37–39]. Shortly, the emulsion samples of 20–40 mg were loaded and then hermetically cold sealed in a DSC aluminum (Al) crucible of 40 μl by volume. Weighing of emulsion samples before and after measurements showed no weight change due to H_2O evaporation. The HL-matrix vitrifies at ~123 K [36] and, consequently, does not perturb the glass transitions of CA/H_2O which are observed between ~176 and 220 K [32,33]. We also used an optical cryo-microscope (OC-M) Olympus BX51 equipped with a Linkam cold stage, Linksys32 temperature control and video capture software for the *in situ* observation and recording of the freezing process and ice/FCS morphology of emulsified drops. To this end we placed emulsions in between a standard 75 × 25 mm microscope slide and a cover glass. OC-M measurements were performed at the cooling/warming rate of 3 and 5 K/min between 300 and 193 K [32,33].

The main body of DSC and OC-M measurements was performed on emulsions in which the diameter of drops varied between the smallest observable size of ~1 μm and the largest size of ~5 μm. The freezing process and ice/FCS morphology of such drops are obscured by the HL-matrix and resolution limit imposed by light wavelength. In order to better understand how ice/FCS morphology change with decreasing drop size, we also employed emulsions in which the diameter of largest drops was ~200 μm and larger. Such emulsions were prepared by varying the rate and time of magnetic stirring.

3. ICE/FCS morphology of frozen bulk CA/H_2O

In this section, we shortly recall the low-temperature behavior and ice/FCS morphology of bulk CA/H_2O because this information is necessary for better understanding the low-temperature behavior and ice/FCS morphology of emulsified CA/H_2O. Concentrated solutions (more than ~65 wt% CA) do not freeze upon cooling and warming [33]. Instead they undergo liquid-glass (CA/H_2O-glass) and reverse glass-liquid transitions upon cooling and warming, respectively. Upon cooling less concentrated solutions (~62–64 wt% CA), increasing viscosity does not prevent the appearance of numerous ice nucleation events [33], but completely suppresses the growth of tiny ice crystals i.e., incipient freezing process is terminated by a CA/H_2O-glass transition. This terminated freezing process resumes upon subsequent warming above a reverse glass-CA/H_2O transition. The solutions of concentration between ~58 and 62 wt% CA freeze upon both cooling and warming [33]. The solutions of concentration less than ~56 wt% CA freeze only upon cooling. FIPS, which occurs during the freezing of CA/H_2O, produces a mixed-phased system: an ice framework (IF) and two freeze-concentrated solutions, FCS_1 and FCS_2 [32,33]. Because of its unique molecular/CA-nanocluster structure, CA/H_2O is very convenient for the *in situ* observation of freezing process and ice/FCS morphology using OC-M. Fig. 1 displays the OC-M images of frozen bulk CA/H_2O solutions of different concentrations. It is seen that IF is entangled with FCS_1 which is *maximally* freeze-concentrated, ~81 wt% CA. Image 1d shows that the IF/FCS_1 as a whole is enveloped by FCS_2 which is a less concentrated, ~75 wt% CA [32,33]. Images 1a–1d show how IF/(FCS_1 + FCS_2) morphology changes with increasing solution concentration.

4. Results and discussions

4.1. The comparison of bulk and emulsion thermograms

In Fig. 2a, we present typical cooling and warming DSC thermograms of bulk CA/H_2O [32,33]. The cooling and warming thermograms of emulsified 10–60 wt% CA are displayed in Fig. 2b. As will be seen below, the comparison of bulk and emulsion thermograms is useful for better understanding the ice/FCS morphology of frozen emulsified drops. Fig. 2a shows that the bulk cooling thermogram contains one exothermic event T_f which is due to the freezing out of pure ice [32,33]. Upon subsequent warming, the melting of ice produces a prolonged endothermic peak T_m with a long low-temperature tail. In DSC method, the number of freezing events usually corresponds to the number of melting events. However, in Fig. 2b, emulsion 10–50 wt% CA cooling thermograms contain two freezing events T_{f1} and T_{f2}, whereas the corresponding

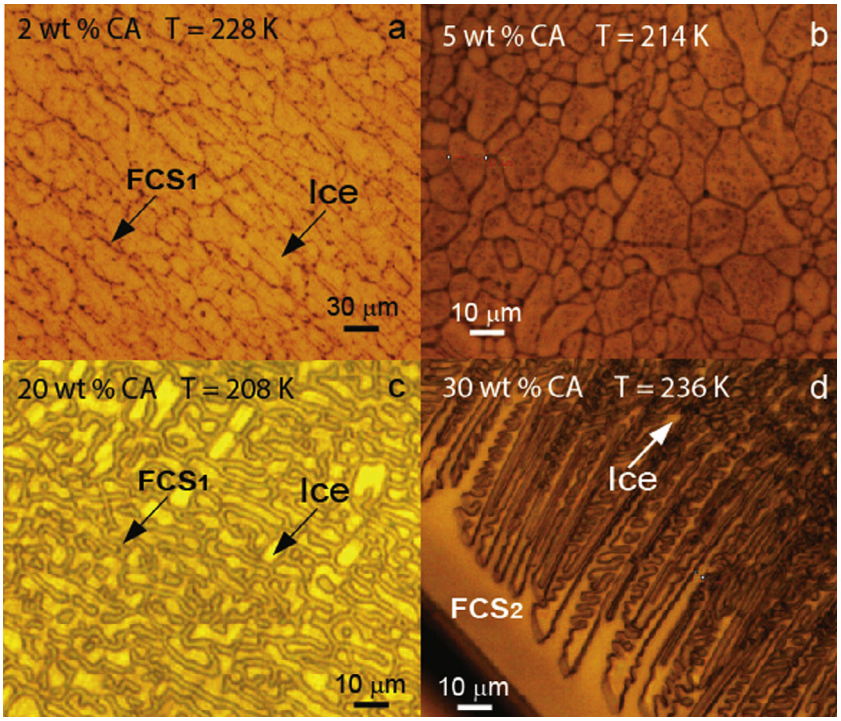


Fig. 1. Optical cryo-microscope (OC-M) images of frozen bulk CA/H_2O show how ice/(FCS_1 + FCS_2) morphology changes with increasing concentration (see text). Concentration, temperature and scale bar are indicated.

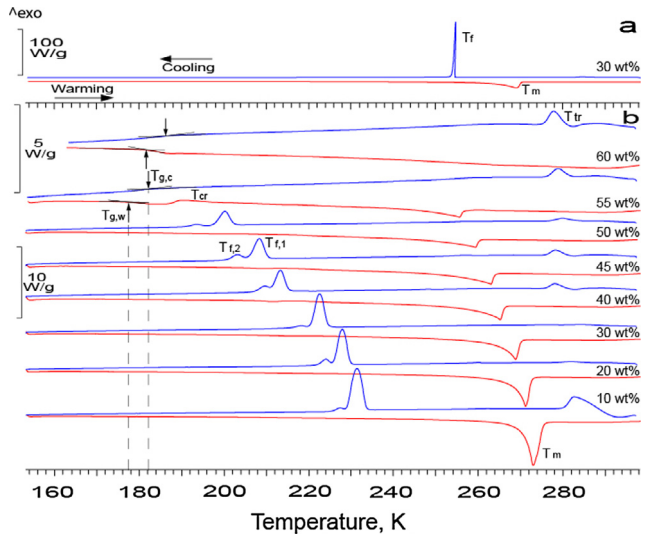


Fig. 2. Cooling (upper blue lines) and warming (lower red lines) thermograms of bulk 30 wt% CA (a) and emulsified 10–60 wt% CA (b). The diameter of drops in emulsions is less that ~5 μm. In panel (a), the freezing and melting of ice are marked by T_f and T_m, respectively. In panel (b), $T_{f,1}$, $T_{f,2}$ and T_{tr} denote two freezing events and CA-CAM transition, respectively (see text). $T_{g,c}$, $T_{g,w}$ and T_{cr} denote a liquid-glass transition, reverse glass-liquid transition and freezing upon warming, respectively. Concentration and the scale-bar of heat flow are indicated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

warming thermograms contain only one ice melting event T_m. In our past works, we reported that the cooling thermograms of emulsified H_2SO_4/H_2O [37,38] and bulk $(NH_4)_2SO_4/H_2O$ [39] contained three and two freezing events, respectively. The warmest freezing event was always due to the freezing out of pure ice. The colder freezing event(s) was/were due to the freezing/crystallization of eutectic mixtures of ice/$H_2SO_4{\cdot}4H_2O$, $H_2SO_4{\cdot}8H_2O$, and ice/$(NH_4)_2SO_4$ respectively. Upon warming, these solids produced endothermic melting peaks at different temperatures. Similarly, in Fig. 2b, we attribute peak $T_{f,1}$ to the freezing out of pure ice. The bell-shaped form of peak $T_{f,1}$ is characteristic for the freezing out of ice in solution-in-oil emulsions because it is formed by the latent heat of fusion released during the freezing of numerous size-distributed drops (>10^5) [36]. About the second freezing event $T_{f,2}$ has not been reported before. The facts that its form is also bell-shaped and the warming thermograms contain only one endothermic melting event T_m suggest that the peak $T_{f,2}$ is produced by the freezing out of pure ice. However, the existence of two ice freezing events would assume that our emulsions contain bimodal size-distributed drops which, in fact, we did not observe in our optical microscope measurements. At the moment, we cannot account for the nature of peak $T_{f,2}$. We are going to perform additional measurements to find out its nature.

In Fig. 2b, emulsion 55 wt% CA thermogram shows that drops do not freeze upon cooling, but undergo a liquid-glass transition $T_{g,c}$. Upon subsequent warming, glassy 55 wt% CA transforms to a highly viscous liquid (HVL) [38] during a reverse glass-liquid transition $T_{g,w}$. Upon further warming, after the viscosity of HVL has decreased, 55 wt% CA solution freezes to produce an exothermic peak T_{cr}. The emulsified solutions of concentration 60 wt% CA and larger do not freeze neither upon cooling or warming. Instead, they produce only liquid-glass and reverse glass-liquid transitions $T_{c,g}$ and $T_{w,g}$, respectively (Fig. 2b). Compared to bulk solutions such 'critical' concentration, at which only liquid-glass and reverse glass-liquid transitions occur, is ~65 wt% CA [33].

Fig. 2b shows that above ~278 K all emulsion cooling thermograms contain an exothermic event T_{tr} which is thought to be due to the transition from CA to CA-monohydrate (CAM) [40,41]. In 20 and 30 wt% CA thermograms, the area of peak T_{tr} is much smaller than that in other thermograms (see also Fig. 3b). This suggests that the CA-CAM transition may be concentration-dependent. Fig. 3a shows that the CA-CAM transition is also present in the cooling thermogram of bulk 30 wt% CA. However, our numerous

DSC measurements suggest that the CA-CAM transition may be more pronounced in emulsified than in bulk solutions. Whether the small size of drops and/or the HL-matrix (i.e., chemical environment) favor the CA-CAM transition is not investigated in this paper.

4.2. The comparison of magnified bulk and emulsion cooling thermograms

Magnified thermograms in Figs. 3 and 4 reveal thermal events not visible in Fig. 2. Fig. 3a shows that the magnified bulk cooling thermogram contains two subtle baseline steps $T_{g1,c}$ and $T_{g2,c}$ which are due to the FCS_1-glass and FCS_2-glass transitions, respectively [32,33]. The onset of these liquid-glass transitions is constant and independent of initial solution concentration $T_{g1,c} \approx 220$ and $T_{g2,c} \approx 208$ K and, consequently, can be considered as the *characteristic* points of CA/H_2O [32,33]. In contrast, Fig. 3b shows that in emulsion cooling thermograms, these two baseline steps are not visible. However, their absence does not mean that FIPS does not occur during the freezing of emulsified CA/H_2O drops. The analysis of emulsion warming thermograms in next paragraph will show that instead of two FCS_1 and FCS_2, which are formed during the freezing of bulk solutions, the freezing of drops smaller than ~5 μm in diameter may produce only one FCS′. Finally, in Fig. 3b, open arrows point out subtle exothermic events T_{exo} whose nature will be discussed in Section 4.5.

4.3. The comparison of magnified bulk and emulsion warming thermograms

Fig. 4 displays magnified warming thermograms from Fig. 2. An additional bulk 50 wt% CA thermogram is presented in Fig. 4a in order to demonstrate that the appearance of bulk warming thermograms is independent of concentration (see also Fig. 4 in Ref. [33]). It is seen that the bulk warming thermograms contain a reverse glass-FCS_2 transition at $T_{g2,w} \approx 204$ K and the T_{tr2}-transition [32,33]. In past these two transitions were called by the T_{tr1} and T_{tr2} transitions, respectively. They have been calorimetrically observed upon warming of frozen carbohydrate solutions

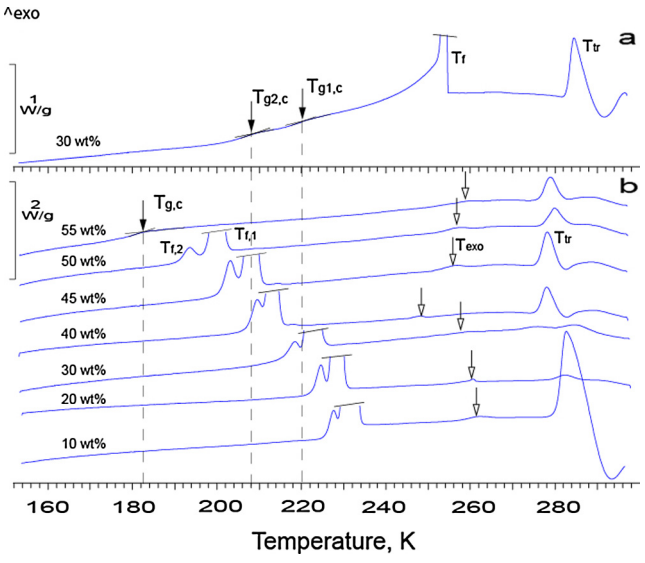


Fig. 3. Magnified cooling thermograms of bulk 30 wt% CA (a) and emulsified 10–55 wt% CA (b) from Fig. 2. $T_{g1,c}$ and $T_{g2,c}$ denote FCS_1-glass and FCS_2-glass transitions, respectively [32,33]. Open arrows mark subtle exothermal peaks T_{exo} which are discussed in Section 4.5. Small inclined lines cut peak $T_{f,1}$ in order to fit figure. Remaining symbols have the same meaning as in Fig. 2.

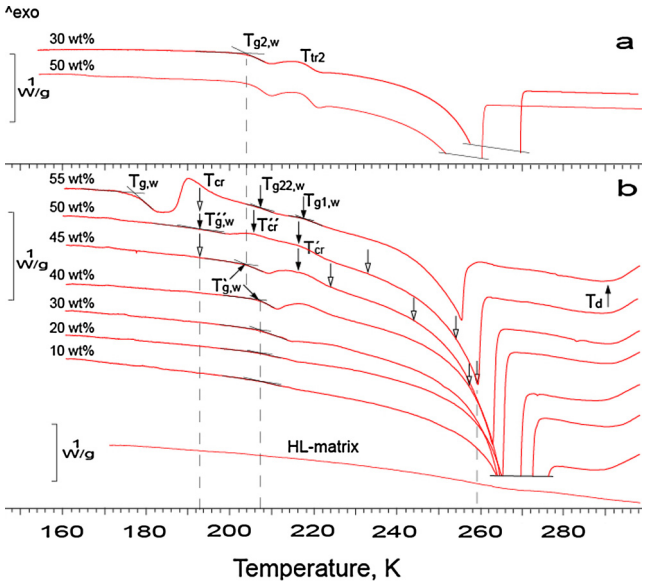


Fig. 4. Magnified bulk 30 and 50 wt% CA (a) and emulsion 10–55 wt% CA (b) warming thermograms. The lowest line in panel (b) is the thermogram of pure HL-matrix. $T_{g,w}$, $T'_{g,w}$, $T''_{g,w}$ and $T_{g22,w}$ mark reverse glass-CA/H_2O, glass-FCS′, glass-FCS″, and glass-FCS_{22} transitions, respectively (see text). Exothermic peaks T'_{cr} and T''_{cr} are due to the freezing upon warming of glassy FCS′ and FCS″. A broad endothermic peak T_d is due to the dissolution of crystalline CA. Small inclined lines cut the ice melting peaks T_m in order to fit figure. Open arrows denote temperatures at which OC-M images displayed in Figs. 6 and 10 were taken.

[42–45]. The cold T_{tr1}-transition was related to a glass-liquid transition. However, the nature of the T_{tr2}-transition, which does not demonstrate a genuine glass–liquid transition baseline step because of a pronounced exothermic feature, has remained a subject of debate for decades. Recently, we showed that the T_{tr2}-transition is a net thermal effect produced by the resumed freezing of FCS_2 and reverse glass-FCS_1 transition at $T_{g1,w} \approx 217$ K. Similar to the onset temperatures of $T_{g2,c}$ and $T_{g1,c}$, the onset temperatures of $T_{g2,w}$ and $T_{g1,w}$ are constant and independent of initial solution concentration and, therefore, can also be considered as the characteristic points of CA/H_2O. The form of the T_{tr2}-transition is determined by (i) the amount of enthalpy emitted during the resumed freezing of FCS_2 i.e., by the amount of FCS_2 and (ii) the height of baseline step and depth of the enthalpy relaxation hollow of glass-FCS_1 transition $T_{g1,w}$ i.e., by the amount of FCS_1 [32,33]. In Fig. 4a, approximately similar height of the baseline steps of $T_{g2,w}$ and $T_{g1,w}$ suggests that the amount of FCS_2 and FCS_1 in frozen bulk solutions is comparable. However, this is not the case of frozen small drops, as is discussed in next paragraph.

The comparison of Fig. 4a and b shows that emulsion warming thermograms differ from bulk warming thermograms. The difference may reflect the difference between the ice/FCS morphology of frozen bulk and emulsified solutions. Fig. 4b shows that emulsion thermograms contain a glass-CA/H_2O transition $T'_{g,w}$ and an exothermic event marked by T'_{cr}. Here, we used symbols $T'_{g,w}$, and T'_{cr} in order to distinguish them from $T_{g,w}$ and T_{cr} in 55 wt% CA warming thermogram (see above). The presence of $T'_{g,w}$ indicates that the corresponding CA/H_2O-glass transition $T'_{g,c}$ should take place upon cooling. However, in cooling thermograms, it is either very subtle or concealed by the two freezing events (Fig. 3b). In contrast to the T_{tr2}-transition in Fig. 4a, the peak T'_{cr} does not contain a baseline step and enthalpy relaxation hollow of $T_{g1,w}$. This indicates that in the frozen drops smaller than ~5 μm in diameter, FCS_1 can be either absent or its amount is very small. This in turn suggests that (i) the ice structure of such drops

is much simpler than that of frozen bulk solutions (Fig. 1) and (ii) this simple ice structure can be coated by a *single* FCS′. Thus, from the above said, we relate $T'_{g,w}$ to a glass-FCS′ transition and T'_{cr} to the subsequent resumed freezing of FCS′. In 10–40 wt% CA warming thermograms, $T'_{g,w}$ is ~3 K warmer than the $T_{g2,w}$ of bulk solutions. This means that the concentration of FCS′ is slightly higher than FCS_2 concentration i.e., slightly higher than ~75 wt% CA [32,33]. In 45 wt% CA thermogram, $T'_{g,w}$ is almost the same as $T_{g2,w}$ of bulk solution and, consequently, the concentration of FCS′ is also ~75 wt% CA. In 50 wt% CA thermogram, $T'_{g,w}$ is not visible because it is concealed by a new exothermic event T''_{cr} (see next paragraph). We can roughly estimate the FCS′ concentration of frozen 50 wt% CA drops from the position of peak T'_{cr}. Since it is similar to that of T'_{cr} in 45 wt% CA thermogram, the concentration of FCS′ of frozen emulsified 50 wt% CA may be also ~75 wt% CA. In 10–20 wt% CA thermograms, very subtle and prolonged $T'_{g,w}$ and T'_{cr} can be due to the very small amount of FCS′ formed during the freezing of *small* and *diluted* drops.

In Fig. 4b., 50 wt% CA thermogram contains a new exothermic peak T''_{cr} which is absent in other thermograms. Undoubtedly, it is also due to ice crystallization because subsequent warming does not reveal a new endothermic melting event. We propose the following mechanism of the formation of peak T''_{cr}. In 55 and 50 wt% CA warming thermograms, exothermic peaks T_{cr} and T'_{cr} are formed by freezing 55 wt% CA and ~75 wt% CA solutions, respectively (see above). The fact that T''_{cr} is in between T_{cr} and T'_{cr} indicates that it should be formed by the freezing upon warming of glassy CA/H_2O whose concentration is in between 55 wt% CA and ~75 wt% CA. In other words, peak T''_{cr} can be formed by the resumed freezing of a new FCS″ whose freezing upon cooling was terminated by a FCS″-glass transition. From our CA/H_2O state diagram (not shown) we estimate that FCS″ concentration can be around ~68–70 wt% CA. Naturally, a following question arises: how come can the FCS″ concentration be smaller than that of FCS′? An answer can be found in 55 and 50 wt% CA *cooling* thermograms in Fig. 2b. Judging from the freezing and glass transition behavior of these two solutions, we assume that emulsified 50 wt% CA sample contains the rare drops of concentration larger than 50 wt% CA, but less than ~55 wt% CA. Upon cooling, such drops would not freeze *completely* but only partially. (Such drops need to possess concentration less than ~55 wt% CA because otherwise they would completely transform to glass upon cooling and upon subsequent warming would produce an exothermic peak similar to T_{cr} in 55 wt% CA thermogram.) For example, such more concentrated drops could be formed owing to the tendency of CA/H_2O to form CA-nanoclusters [4–9]. During the freezing of these rare more concentrated drops, propagating ice front pushes ahead FCS′ which will mix with unfrozen initial CA/H_2O and produce FCS″. (How propagating ice/FCS_1 front pushes ahead unfrozen CA/H_2O is demonstrated in Movie 2 in SI of Ref. [32].) Because the size of drops is less than ~5 μm, the amounts of FCS′ and unfrozen CA/H_2O in each drop are very small and, therefore, they may mix completely at the experimental cooling rate of 3–5 K/min. Upon further cooling, the FCS″-glass transition will terminate freezing and, consequently the drops would remain only *partially* frozen. Upon subsequent warming above a reverse glass-FCS″ transition $T''_{w,g}$, the terminated freezing would recommence and produce peak T''_{cr}. As a confirmation of this reasoning can serve a prolonged glass-FCS″ transition in Fig. 4b with the onset at $T''_{w,g} \approx 193$ K which, according to our CA/H_2O state diagram, corresponds to the concentration of ~68 wt% CA. The prolonged $T''_{w,g}$ is due to the distribution of FCS″ concentration in different drops. In Section 4.4, we will present the OC-M images of drops larger than 5 μm in diameter which show that *partly* frozen drops indeed exist in emulsified 50 wt% CA. These images can be considered as another confirmation of the proposed mechanism of the formation of peak T''_{cr}.

Farther, there is a surprising similarity between emulsion 55 wt% CA warming thermogram in Fig. 4b and the warming thermogram of *quenched* bulk 55 wt% CA in Fig. 9 of Ref. [33]. In both these thermograms, the onset of glass-CA/H_2O transition is at $T_{g,w} \approx$ 177 K, the temperature regions of HVL and peak T_{cr} are similar, and the T_{tr2}-transition is absent. What is more, emulsion 55 wt% CA warming thermogram also contains two glass-liquid transitions, $T_{g22,w}$ and $T_{g1,w}$, at *exactly* the same temperatures as those in quenched bulk 55 wt% CA warming thermogram. For the time being we cannot account for this remarkable similarity between quenched bulk and emulsion 55 wt% CA warming thermograms.

Finally, in Fig. 5, we compare ice melting peaks T_m in bulk and emulsion warming thermograms reduced to a single scale. It is seen that the form of both peaks is quiet similar i.e., they both contain a long low-temperature tail. The existence of long low-temperature tail indicates that ice starts melting at an ice/highly-concentrated-solution interface. We know that upon warming of frozen bulk solutions, ice starts melting from the ice/FCS_1 interface at ~220 K and that FCS_1 is maximally freeze-concentrated [33]. Fig. 5 shows that above this temperature the two low-temperature tails are quiet 'parallel'. This parallelism suggests that upon warming of the frozen drops of diameter smaller than ~5 μm, ice starts to melt also from the ice/FCS_1 i.e., upon warming the resumed freezing of FCS′ and FCS″ may also produce the maximally freeze-concentrated FCS_1. In emulsion 55 wt% CA warming thermogram in Fig. 4b, the existence of $T_{g1,w}$ confirms that indeed the freezing upon warming produces the maximally freeze-concentrated FCS_1.

4.4. OC-M images of frozen emulsified CA/H_2O

In Fig. 6, we present OC-M images of frozen emulsified 50 wt% CA which confirm our suggestion posed above that the ice structure of drops smaller than ~5 μm in diameter is less ramified than that of frozen bulk solutions. The images are taken at temperatures indicated by open arrows in Fig. 4b. Note that the location of image 6a differs from that of images 6b–6d. In our OC-M measurements, freezing drops are detected by fast feeble darkening (see Movie 1 in Ref. [46]). The darkening during freezing is better seen for larger

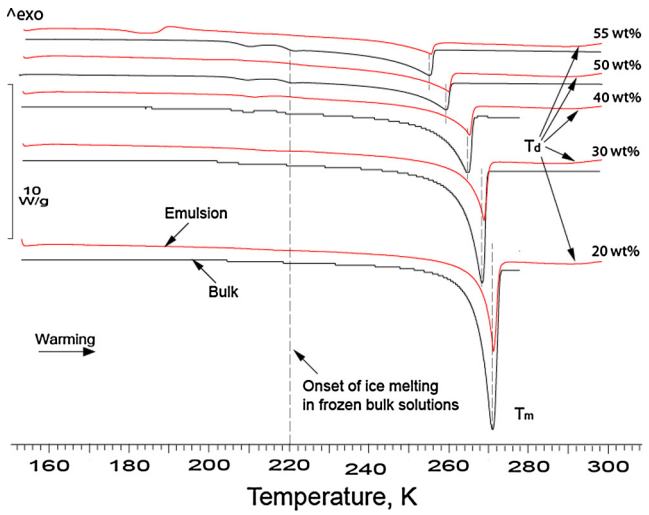


Fig. 5. Superposition of the warming thermograms of emulsified CA/H_2O (upper red lines) and bulk solutions (lower black lines). The warming thermograms are taken from Fig. 4b of this work and Fig. 4 in Ref. [33] and reduced to a single scale. Short dashed lines show the shift of T_m of emulsified CA/H_2O to warmer temperature with respect to the T_m of bulk solutions. Arrows show a broad endothermic peak T_d in emulsion thermograms. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

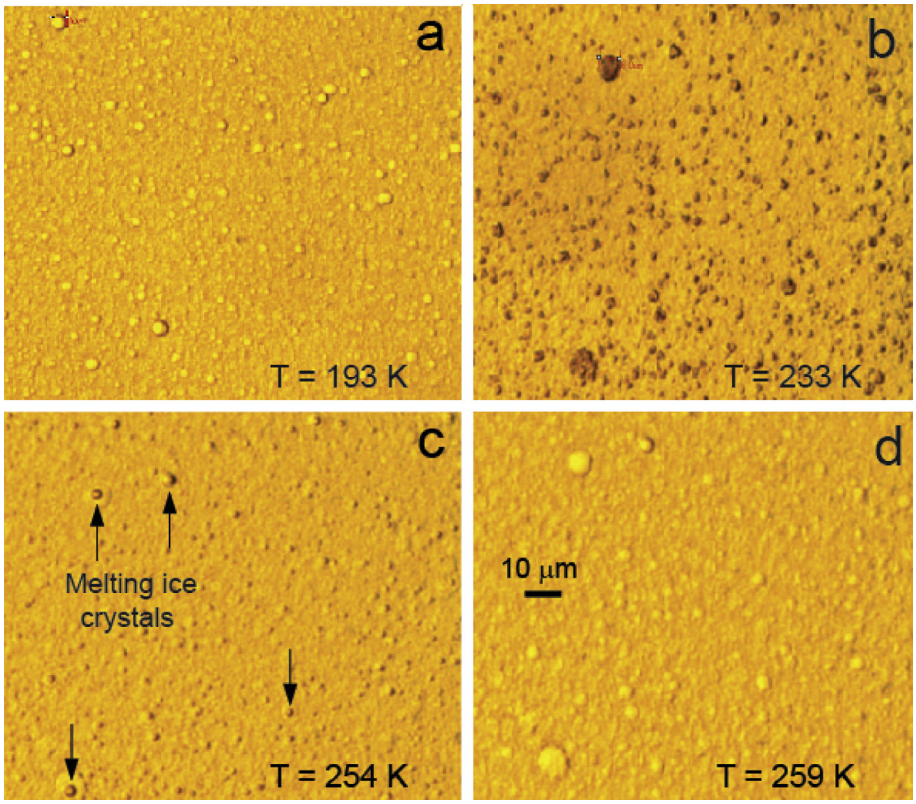


Fig. 6. OC-M images obtained upon warming of frozen 50 wt% CA drops of diameter smaller than ~5 μm. Images are taken at temperatures indicated by open arrows in 50 wt% CA thermogram in Fig. 4b. In image 5c, arrows show remaining melting ice crystals. In image 6d, all ice has melted. A scale bar of 10 μm is applied to all images.

drops (see Movie 2 in Ref. [46]). Images 6a and 6f show that frozen and unfrozen small drops are almost not distinguishable i.e., the ice structure of frozen small drops is not observable because it is concealed by the double-layer consisting of glassy FCS′ and HL-matrix. However, ice structure can approximately be revealed upon warming, as is seen in images 6b. Above the glass-FCS′ transition $T'_{g,w}$, the resumed freezing depletes FCS′ layer and, at sufficiently high temperature, ice structure becomes more or less distinguishable (see Movie 3 in Ref. [46]). Images 6b and 6c show that frozen drops smaller than ~2–3 μm in diameter contain a quasi-spherical or shapeless crystal surrounded by a single FCS′. The ice structure of drops larger that ~3–4 μm in diameter becomes more complex and, as drop diameter increases, approaches that of frozen bulk solutions (see images below). Unfortunately, the exact ice structure of drops smaller than ~5 μm is difficult to observe in OC-M measurements because of limitations imposed by light wavelength and due to the presence of several different optical media such as FCS, the HL-matrix, cover glass, air, glass of a cold chamber, and again air.

Fig. 7 presents the OC-M images of frozen 50 wt% CA drops smaller than ~50 μm. The images are taken upon cooling at 193 K i.e., below $T_{f,1}$ in 50 wt% CA cooling thermogram. Upon cooling of large 50 wt% CA drops, besides rapidly frozen drops, we also detect some drops in which freezing process is sluggish and ice structure is more or less observable. The freezing of such drops is often initiated from multiple ice nucleation events (see Movie 4

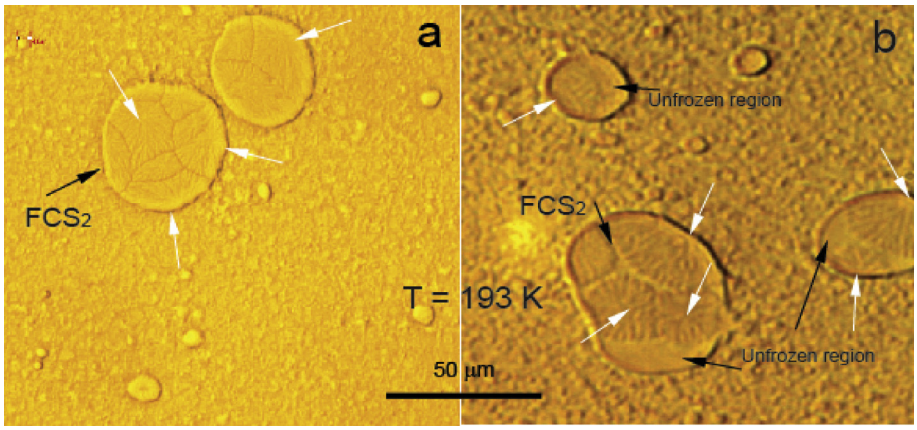


Fig. 7. OC-M images taken at 193 K after the freezing upon cooling 50 wt% CA drops of diameter less than 50 μm. Black arrows mark the channels of FCS_2 and unfrozen CA/H_2O regions. White arrows mark the location of ice nucleation events from which spherulitic ice structure has been formed. Scale bar is indicated.

in Ref. [46]). For example, in image 7a, the freezing of the largest drop is initiated from 8 ice nucleation events which produce 8 spherulitic ice structures. We know from the OC-M study of bulk CA/H_2O that multiple ice nucleation events and sluggish freezing are characteristic of the freezing solutions of concentration larger than ~55 wt% CA [32,33]. Consequently, the concentration of our drops can also be larger than ~55 wt% CA. Further, images 7a and 7b show that spherulitic ice regions are filled with FCS_1, as in frozen bulk solutions (Fig. 1). Image 7b also shows that, as the ice phase grows, propagating ice/FCS_1 front pushes ahead FCS_2 which envelops the whole ice/FCS_1 structure and fills the lines of the intersection of spherulitic ice regions, as is seen in image 7a. Thus images in Fig. 7 demonstrate that large frozen drops possess IF/(FCS_1 + FCS_2) morphology similar to that of frozen bulk solutions.

In Fig. 8, we present additional OC-M images which demonstrate that large frozen drops possess IF/(FCS_1 + FCS_2) morphology similar to that of frozen bulk solutions. The images are taken at 203 K upon cooling of emulsified 30 wt% CA. The different brightness of frozen drops is due to their different freezing temperatures: darker drop freeze at temperature warmer than that of brighter drops. Since the size of drops is comparable, the different freezing temperatures may stem only from different drop concentrations. More about concentration-distributed drops in emulsions will be said below.

Further, image 7b shows that three large drops are partly frozen. We do not know their exact concentration, because we did not record their resumed freezing upon warming and subsequent ice melting during our OC-M measurements. Drop concentration can be estimated from ice melting peaks T_m in warming thermograms (Fig. 2b). All what we can say about the concentration of these partly frozen drops is that it is larger than the concentration of drops in image 7a i.e., larger than 55 wt% CA. Instead, in Fig. 9, we present the sequence of OC-M images taken during the warming of another partly frozen drop (see Movie 5 in Ref. [46]). Image 9a shows that upon previous cooling, freezing had been heterogeneously initiated by (or near) a CA crystal formed at the HL-matrix/solution interface. To our best knowledge, there have been no reports about the crystallization of CA or CAM during the cooling of emulsified CA/H_2O. In this work, we assume that CA crystallizes upon cooling. Image 9b shows that, as temperature increases, a branched ice structure grows due to the resumed freezing of FCS_2 and/or initial CA/H_2O. In image 9c, the growing ice structure reaches its maximum at ~230 K. It is interesting that this temperature exactly matches the peak temperature T_{cr} in *bulk* 62 wt% CA warming thermogram in Fig. 1 in Ref. [16]. Above 230 K, ice intensively melts and the continuous ice structure disintegrates into numerous isolated ice crystals (image 9d). Image 9e shows that complete ice melting occurs at ~248 K. This temperature corresponds to the peak melting temperature $T_m \approx 248$ K in bulk

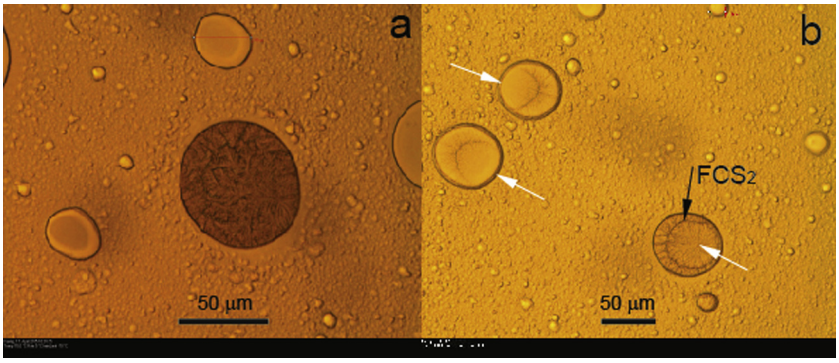


Fig. 8. OC-M images taken at 203 K upon cooling emulsified 30 wt% CA demonstrate different brightness of frozen drops. The darker drops froze at warmer temperature. Black arrow marks the channels of FCS_2. White arrows mark the location of ice nucleation events.

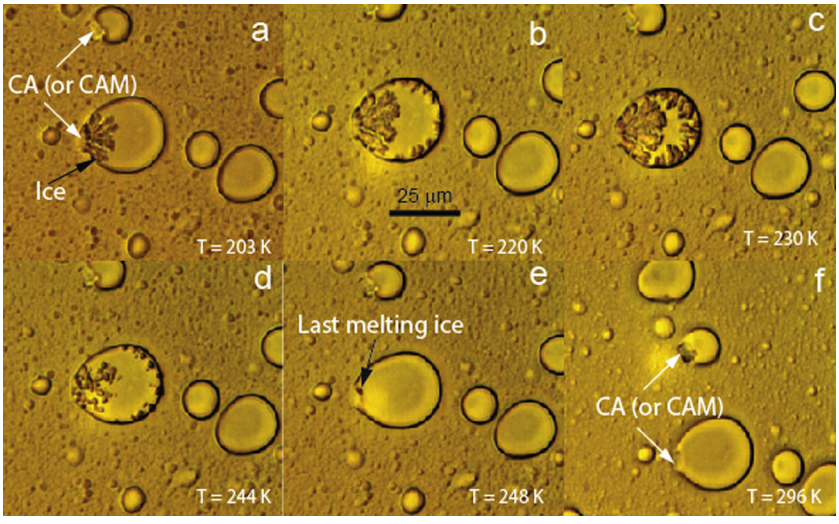


Fig. 9. The sequence of OC-M images taken upon warming of a partly frozen concentrated drop. White arrows show CA crystals which induced heterogeneous ice nucleation and subsequent freezing. In image 9a, a black arrow shows ice structure 'frozen-in' by a CA/H_2O-glass transition upon cooling. In image 9e, a black arrow marks a dark spot which is a last melting ice crystal. A scale bar of 25 µm is applied to all images.

62 wt% CA warming thermograms [33], suggesting that the concentration of this particularly drop is ~62 wt% CA. The brightness of other, smaller drops in Fig. 9 does not change upon warming. This means that they undergo only CA/H_2O-glass and reverse glass-CA/H_2O transitions. Unfortunately, unlike freezing, which makes drops more or less darker depending on the drop size, CA/H_2O-glass and reverse glass-CA/H_2O transitions do not change drop brightness. Therefore OC-M images cannot distinguish between completely glassy drops and *freshly* frozen small drops enveloped with glassy FCS′. As we mentioned above, the concentration of bulk solutions, which do not freeze upon cooling and warming, is larger than ~64 wt% CA (see also Fig. 1 in Ref. [33]). Note that the 'critical' concentration decreases with solution volume. For example, for drops <5 μm in diameter it is ~55 wt% CA (Fig. 2b).

In contrast to concentrated large drops, whose images have been discussed in Figs. 7–9, the ice structure of less concentrated drops cannot usually be observed after freezing upon cooling (see Movie 6 in Ref. [46]). Fortunately, it can be partly revealed upon warming of frozen drops, as has been demonstrated by the OC-M images of small drops in Fig. 6. In Fig. 10, we present the OC-M images of frozen 45 wt% CA drops which demonstrate how ice structure reveals itself with increasing temperature. The images are selected from Movie 7 [46] at temperatures indicated by open arrows in 45 wt% CA warming thermogram in Fig. 4b. Analyzing OC-M images, it is useful to refer to the corresponding thermograms because it gives us information about the phase state of drops. Image 10a taken at ~193 K shows that drops froze at different temperatures because their brightness is different: larger darker drops froze at temperature warmer than smaller bright drops marked by arrows. In image 10a, ice structure is not visible because it is concealed by glassy FCS_2 and HL-matrix. Image 9b was taken at 224 K when the resumed freezing of FCS_2 finished and ice, which is in contact with the maximally concentrated FCS_1, started to melt [32,33]. At this temperature the amount of ice is maximal and all drops, including smallest ones, become darker because more light is absorbed by ice/FCS_1 than that absorbed by glassy FCS_2 or glassy CA/H_2O. Image 10c was taken at 244 K when ice melting is in progress and, consequently, the concentration of FCS_1 decreases. Here, ice structure reveals itself as numerous bright spots which are ice crystals in contact with a cover glass. Image 10d taken at 257 K shows that ice structure has been disintegrated into numerous isolated ice crystals floating in diluted FCS_1. We observed that the majority of drops completely melted at ~ 263 K which is similar to $T_m \approx 263$ K in 45 wt% CA warming thermogram (Fig. 2b). We also observed the complete ice melting of some drops at temperature colder than $T_m \approx 263$ K that indicates that their concentration is larger than 45 wt% CA.

In Fig. 11, we present OC-M pictures which show that the different brightness of frozen drops is due to different concentrations. The exact freezing temperature of these drops was not recorded upon cooling. In image 11a, arrows 1 and 2 mark the darkest and brightest drops, respectively. Images 11b–11d are snapshots of the resumed freezing of FCS_2 and subsequent ice melting. Image 11e shows that in the drop 2, the complete ice melting ends at ~263 K. This indicates that drop concentration is ~45 wt% CA (see Fig. 2b). Image 11f shows that in the drop 1, the complete ice melting ends at ~269 K, indicating that its concentration is ~30 wt% CA. The concentration difference as large as 15 wt% CA is surprising, because it is usually believed that all drops in emulsions possess uniform concentration similar to that of initial solution. In the case of CA/H_2O, the different drop concentrations may be accounted for by the tendency of CA/H_2O to form CA-nanoclusters [4–9]. Note that in Fig. 2, the ice freezing peak T_{f1} extends over the temperature region of ~6–7 K. Since drop size distribution is small (they all are smaller than ~5 μm), the extended ice freezing region may be due to the distribution of drop concentration.

Further, in image 11a, an arrow 1 marks a thin dark channel issuing from the frozen drop 1. Similar channels often appear

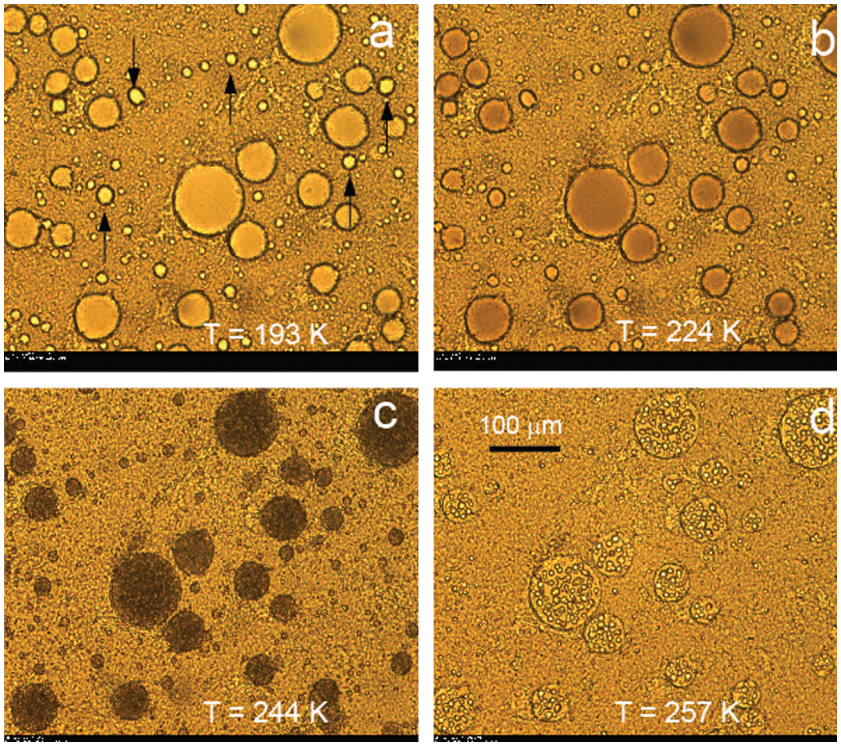


Fig. 10. A sequence of OC-M images taken upon warming of frozen emulsified 45 wt% CA drops. Arrows mark several bright drops which are either glassy or frozen at temperature much colder than the temperature at which large and darker drops froze (see text). In image 10c, the tiny bright spots in partly melted drops are ice crystals in contact with a cover glass. In image 10d, intensive ice melting disintegrated continuous ice structure into numerous isolated ice crystals. A scale bar of 100 μm is applied to all images.

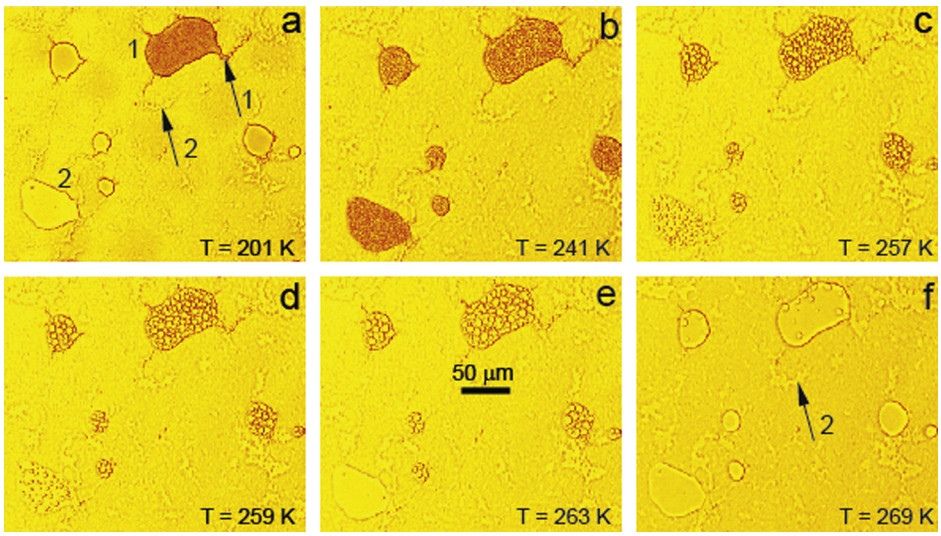


Fig. 11. The sequence of OC-M images obtained upon warming of frozen drops of unknown concentration. The darkest 1 and brightest 2 drops possess the smallest and largest concentration, respectively (see text). Arrow 1 marks a thin dark channel issuing from the frozen drop. Arrow 2 marks some unknown feature in the HL-matrix throughout all images. A scale bar of 50 μm is applied to all images.

during the freezing of emulsified drops (see Movie 2 in Ref. [46]). In Fig. 12, we present two images obtained in transmitted (image 12a) and reflected (image 12b) light modes which demonstrate that the channels issuing during freezing are crystalline. Image 12b also shows that the drops of different brightness look practically similar in the reflected light mode. The origin of channels issuing during the freezing of drops is not completely clear. We hypothesize that in emulsions, drops may experience external pressure from the encircling HL-matrix. If the thermal expansion coefficient of HL-matrix is larger than that of CA/H_2O then, as temperatures decreases, the HL-matrix will squeeze drops. Our OC-M measurements show that during freezing, drops expand and this will further increase pressure on the lanolin/oil bilayer around drops. Resulting pressure may be large enough to break the surfactant/oil bilayer and produce issuing channels. Finally, in Fig. 11, arrow 2 marks a shapeless region in the HL-matrix whose nature is also unclear. These regions often appear during the cooling of emulsions and disappear upon warming when temperature rises above ~273 K (images not shown). Note that in Fig. 12 these regions are not visible.

In Fig. 13, we present OC-M images which demonstrate another interesting situation which often arises during the freezing of emulsified drops. Image 13a shows that in addition to small issuing channels, there are spherical segments at the edge of frozen drops. Image 13b demonstrates that they are not frozen. Most likely the spherical segments are filled with FCS_2 which freezes upon warming.

4.5. The crystallization and dissolution of CA

In Fig. 4b, all emulsion warming thermograms contain a broad endothermic event T_d above 273 K which is absent in bulk warming thermograms (see Fig. 4a and Fig. 4 in Ref. [33]). The first explanation, which comes in mind, is that it could be produced by the HL-matrix. To verify this hypothesis, we performed DSC measurements of the pure HL-matrix at experimental conditions similar to those of emulsion measurements. The obtained warming thermogram of HL-matrix is displayed at the bottom in Fig. 4b. The comparison of emulsion and pure HL-matrix thermograms clearly demonstrates that the peak T_d can't be produced by the HL-matrix. One may suggest that the peak T_d can be due to the dissolution of CA crystals which had crystallized upon previous cooling. For example, CA crystals are clearly seen in the images in Fig. 9. In our OC-M measurements, we indeed observe the dissolution of crystalline CA upon warming of frozen concentrated drops. Exemplary images of CA dissolution are presented in Fig. 14. However, we have never observed CA dissolution upon warming of frozen 10–20 wt% CA drops, although their warming thermograms also contain clearly pronounced peak T_d (Fig. 4b). Nevertheless, it seems that in the diluted drops, undissolved crystalline CA also

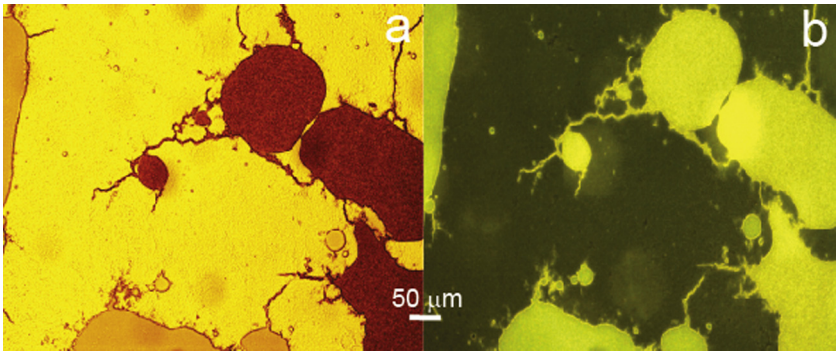


Fig. 12. OC-M images obtained at 203 K upon cooling 25 wt% CA drops. (a) Transmitted light mode and (b) reflected light mode. A scale bar of 50 μm is applied to both images.

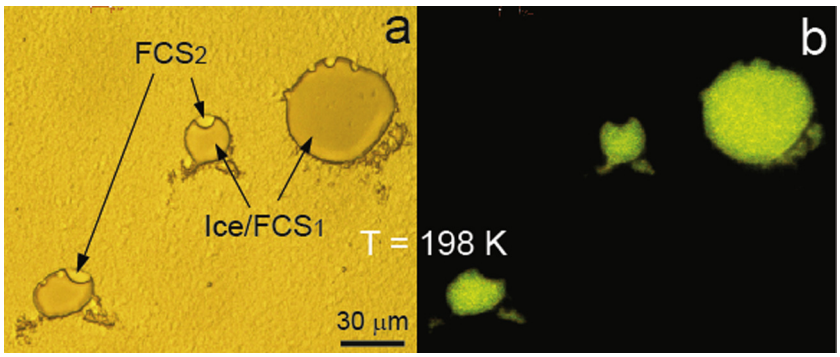


Fig. 13. OC-M images obtained upon cooling emulsified 20 wt% CA in transmitted (a) and reflected (b) light modes at 198 K. Arrows show the FCS_2 and ice/FCS_1 of frozen drops (see text).

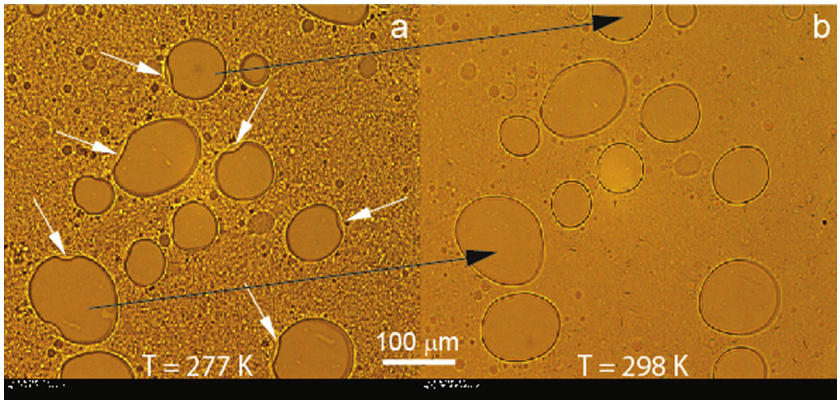


Fig. 14. OC-M images obtained at 277 and 298 K demonstrate the dissolution of crystalline CA upon warming of frozen 45 wt% CA drops. White arrows mark CA crystals. Long black arrows show a pair of the same drops.

exists above 273 K because, as is seen in Fig. 5, short vertical dashed lines show the manor warm-temperature shift of peak T_m in emulsion thermograms with respect to peak T_m in bulk thermograms. The shift increases with concentration from ~0.5 K for 10 and 20 wt% CA to ~1 K for 50 wt% CA that is physically reasonable, because in more concentrated solutions, more CA crystallizes upon cooling and less dissolves upon warming. Thus the peak T_d is indeed due to the dissolution of CA crystals crystallized upon previous cooling.

If warming thermograms contain endothermic dissolution peak T_d then cooling thermograms should contain exothermic CA crystallization peak. One may suggest that in Fig. 3b, small exothermic peaks T_{exo} could be due to CA crystallization. To verify this possibility, in Fig. 15 we compare (i) emulsion and pure HL-matrix cooling thermograms and (ii) the areas of peaks T_{exo} and T_d. Since in the HL-matrix thermogram, peak T_{exo} is absent, we conclude that it can be produced by some exothermic process which occurs in emulsified CA/H_2O. However, the fact that the area of peaks T_{exo} is much smaller than that of peaks T_d indicates that the peaks T_{exo} cannot be produced by CA crystallization. Note that when comparing the areas of peaks T_{exo} and T_d, we take into account the fact that their temperature regions are relatively close to each other and, consequently, the enthalpy change associated with the crystallization and dissolution of CA should be similar. To our best knowledge, there are no reports that the enthalpy change associated with the crystallization and dissolution of CA depends on temperature. Thus the origin of peaks T_{exo} is also unclear. We assume that upon cooling CA crystallization is a slow process which occurs over large temperature region and, therefore, is not detectable in our DSC measurements. In our OC-M measurements of emulsions, we did not observe CA

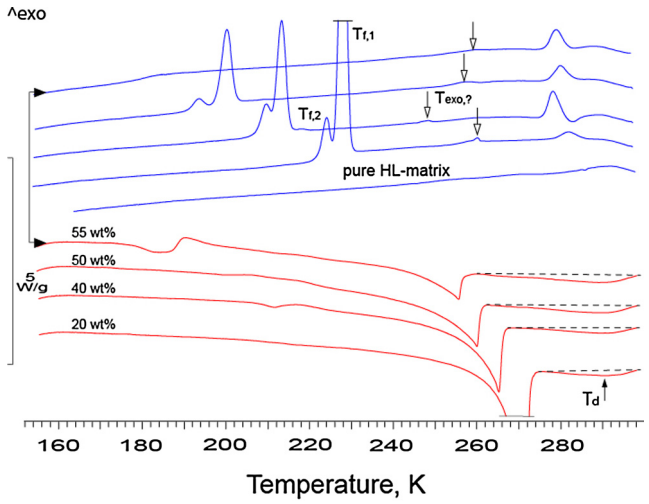


Fig. 15. Selected magnified cooling and warming thermograms from Fig. 2a show that the area of peaks T_{exo} is much smaller than that of peaks T_d. The lowest cooling thermogram was obtained from the pure HL-matrix.

crystallization either. However, upon cooling of bulk concentrated solution CA crystallization can easily be visualized (images are not shown).

In Fig. 16, we present a sequence of OC-M images taken from Movie 8 [46] recorded upon warming of frozen 50 wt% CA drop of diameter ~400 μm. These images demonstrate three different physical processes: (i) the resumed freezing of FCS_2, (ii) ice melting and (iii) dissolution of crystalline CA. The drop in Fig. 16 differs

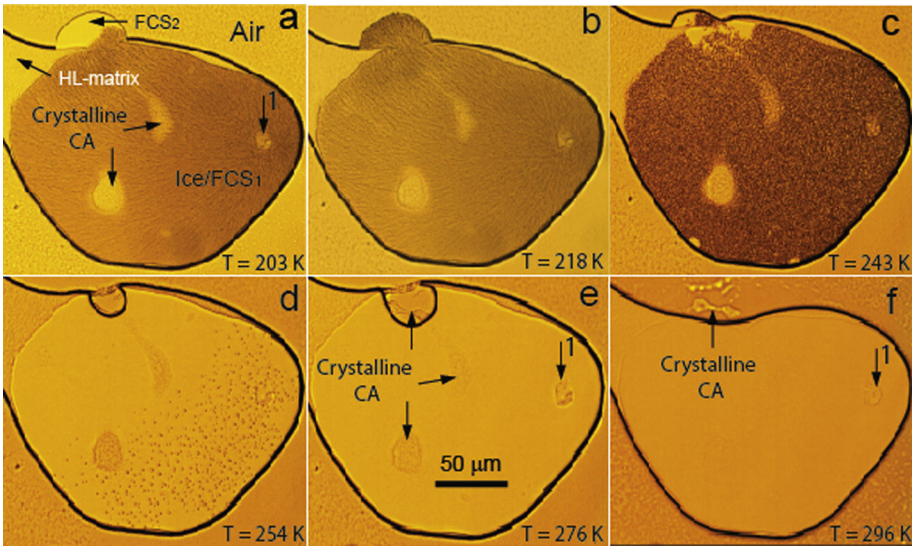


Fig. 16. OC-M images obtained upon warming of frozen 50 wt% CA drop of diameter ~400 µm. Arrow 1 marks a CA crystal from which radial ice twigs/branches propagate in all directions (see text for details). Two other arrows show the bright regions of polycrystalline CA. Scale bar of 50 µm is applied to all images.

from large drops investigated before by the size and large fraction of solution/air interface. Only a small part of drop perimeter borders with the HL-matrix. Image 16a demonstrates that freezing upon cooling was initiated by (or near) CA crystal marked by a vertical arrow 1. This is inferred from radial ice twigs/branches propagating from the CA crystal. Two bright regions marked by other arrows are *polycrystalline* CA. The CA crystal and two CA polycrystalline regions are within the drop; that is, they are not in contact with the solution/lanolin interface. This is in contrast to the OC-M images in Figs. 9 and 14 which show that in drops smaller that ~100 µm in diameter, CA crystalized at the solution/lanolin interface. Whether CA crystallization is heterogeneously induced by the HL-matrix or facilitated by increasing curvature in small drops remains to be investigated.

The freezing temperature of the drop in Fig. 16 was not recorded because upon cooling of the drop our microscope was foced over other place. However, we see in image 16a that during freezing, the propagating ice/FCS_1 front pushed ahead FCS_2 which broke the solution/air interface and made a FCS_2 bulb. Image 16a also shows that the FCS_2 bulb is only partly frozen, because its freezing was terminated by the FCS_2-glass transition. Upon subsequent warming, the freezing of FCS_2 resumes (image 16b). In images 16b–16d, the resumed freezing of FCS_2 and ice melting are in progress. Ice melting ends at ~254 K that indicates that drop concentration is ~56 wt% CA. Further, images 16c–16f show that water evaporates through a rapture in the solution/air interface. The evaporation of water makes the FCS_2 bulb and then all drops shrink. Because of large surface area, the two polycrystalline CA regions dissolve faster than the single CA crystal which remains undissolved even at 296 K. In this context, one may suggest that water evaporation could be a reason of the different drop concentrations discussed above in Figs. 8–11. However, water evaporation cannot explain the concentration difference as larger as 15 wt% CA of drops which are separated by only ~170 µm in Fig. 11. Thus the question of why do emulsions contain drops of different concentrations remains open.

Further, to our best knowledge, image 16a is the first evidence that ice nucleation and subsequent freezing are initiated by (or near) CA crystal. We write "by (or near)" because it is unclear whether *heterogeneous* ice nucleation is induced by CA crystal itself or *homogeneous* ice nucleation occurs near CA crystal in the solution in which concentration was reduced by CA crystallization. The knowledge of whether crystalline CA induces heterogeneous ice nucleation may have implication in the atmosphere, where aqueous aerosol drops containing CA have been observed [35].

Finally, in Fig. 17 we present an OC-M image of a crystalline network which often appears in the HL-matrix when the temperature of emulsions lowers below ~200 K. The network continuously grows as temperature lowers to ~153 K, but never occupies all HL-matrix. Upon warming it melts/disappears when temperature increases to ~210–220 K. Note that in DSC thermograms, there is no indication of exothermic and endothermic events in these temperature regions. This indicates that the appearance and disappearance of the crystalline network are subtle thermal events and not detectable by our DSC. Further, we performed OC-M measurements of pure HL-matrix between 293 and 153 K and found that the crystalline network is practically absent. This fact indicates that there might be some interaction between the HL-matrix and CA/H_2O which is responsible for the appearance of the crystalline network. Nevertheless, the exact nature of its appearance is unclear.

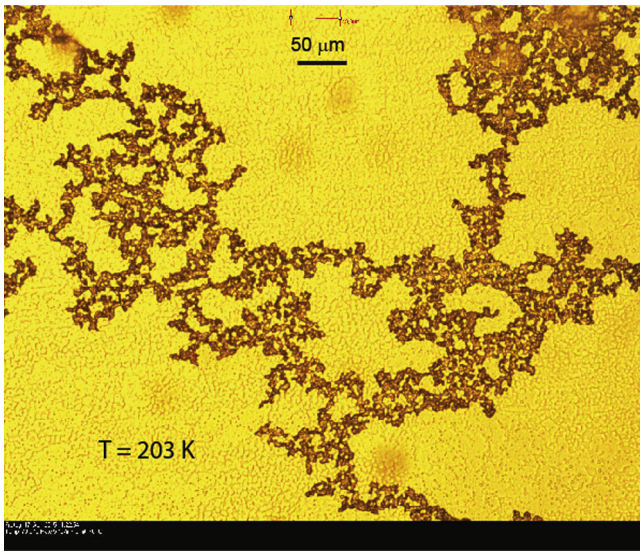


Fig. 17. An OC-M image of a crystalline network in the HL-matrix at 203 K.

5. Conclusions

In this work, we have studied the freezing and glass transitions upon cooling and warming and the ice/FCS morphology of emulsified 10–60 wt% CA solutions using DSC and OC-M techniques. During the study we have obtained a lot of important results which demonstrate that DSC thermograms and OC-M images/movies are mutually supplementary and, consequently, allow us to obtain knowledge which cannot be gained when DSC and OC-M techniques are used alone. Since our results to our best knowledge are first of their kind and, naturally, not all are unambiguous, we have divided them into two groups. In the first group, we collected results which are clear at the moment. In the second group, we collect unclear results which require future investigations.

Clear results: (1) Similar to bulk CA/H_2O, emulsified CA/H_2O solutions also freeze upon cooling and warming. (2) Emulsion warming thermograms contain one glass-FCS′ transition that contrasts the two glass-FCS_1 and glass-FCS_2 transitions of bulk solutions. The concentration of FCS′ is in between the concentrations of FCS_1 and FCS_2 of bulk solutions. (3) The analysis of bulk and emulsion thermograms and the *in situ* OC-M observation of freezing emulsions, show that frozen drops smaller than ~2–3 μm in diameter contain a quasi-spherical or shapeless ice crystal surrounded by a single FCS′. The ice/FCS′ morphology of drops larger than ~3–4 μm in diameter becomes more complex and approaches the ramified IF/(FCS_1 + FCS_2) morphology of bulk solutions when drop diameter reaches ~15–20 μm. (4) 'Critical' concentration, at which CA/H_2O does not freeze upon cooling and warming, but undergoes only CA/H_2O-glass and reverse glass-CA/H_2O transitions, decreases with the size/volume of CA/H_2O. For example, for bulk solutions it is ~65 wt% CA [16] and for drops smaller than ~5 μm in diameter ~60 wt% CA. (5) The 'parallelism' of the low-temperature tails of ice melting peaks T_m in bulk and emulsion warming thermograms suggests that frozen emulsified drops start melting also from the ice/FCS_1 interface. (6) Upon cooling emulsified solutions, the crystallization of CA is slow process which spreads over a large temperature region and, therefore, is not visible in cooling thermograms. In contrast, warming thermograms contain a broad endothermic peak above 273 K which is due to the dissolution of crystalline CA. The dissolution of CA is clearly seen in OC-M measurements.

Unclear results: (1) In contrast to bulk solutions, which produce one freezing event T_f, CA/H_2O drops of diameter smaller than ~5 μm produce two freezing events $T_{f,1} > T_{f,2}$. Although we hypothesize that the colder event $T_{f,2}$ could be due to the freezing of drops smaller than ~1 μm, which are not detectable by our optical microscope, its genuine nature is not clear. (2) In general, bulk and emulsion warming thermograms are different. Therefore, the remarkable similarity between the warming thermograms of *quenched* bulk and emulsion 55 wt% CA is not clear. (3) It is not clear whether the HL-matrix triggers the heterogeneous crystallization of CA or CA-crystallization is facilitated by the increasing curvature of small drops. (4) It is unclear whether CA crystals induce *heterogeneous* ice nucleation or *homogeneous* ice nucleation takes place near CA crystals in the solution region in which concentration was reduced by CA crystallization. (5) The origin of subtle exothermic peaks T_{exo} in cooling thermograms is unclear (Figs. 3b and 15). (6) The physical reason of the existence of drops of different concentrations in emulsions is not known. (7) The appearance of crystalline network in the HL-matrix in emulsions below ~205 K is unclear (Fig. 17).

Our results demonstrate that the combined DSC/OC-M measurements are the valuable source of information about the freezing, glass transitions and ice/FCS morphology of emulsified binary CA/H_2O. The combined DSC/OC-M measurements can be used for the study of bulk and emulsified multicomponent formulations relevant to biotechnology, biopharmaceutics, tissue engineering, food industry, life sciences, etc.

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